BHASKAR AHUJA

Date of birth: 06/03/1998 | **Place of birth:** Fatehabad, India | **Nationality:** Indian | **Gender:** Male |

Phone number: (+39) 3476992692 (Mobile) | **Email address:** bhaskar.ahuja@unitn.it | **Website:** https://orcid.org/0009-0001-8126-2031 | **Address:** Fatehabad, India (Home) |

Address: Via Emilia 41, Pisa, 56124, Pisa, Italy (Current Address)

ABOUT ME

PhD researcher in Space Domain Awareness specializing in passive radar, bistatic tracking, and orbit determination. Experienced in space object tracking using radio telescope receivers, multi-sensor data fusion, and advanced filtering techniques (EKF, UKF). Seeking a postdoctoral position to advance research in radar-based space surveillance and trajectory prediction.

RESEARCH INTERESTS

Space Domain Awareness, Passive Radar, Orbit determination and tracking, Antenna arrays and beamforming, Signal processing for space surveillance

EDUCATION AND TRAINING

01/11/2022 – CURRENT Trento, Italy

PH.D IN SPACE SCIENCE AND TECHNOLOGY University of Trento

- Advanced concepts in astrodynamics, orbit determination, and satellite tracking
- Implementation of signal processing, data fusion, and machine learning techniques for space applications
- Research project planning, academic writing, and dissemination through conferences and journals

Website https://phd.unitn.it/phd-sst/ | **Field of study** Physics | **Level in EQF** EQF level 8 |

Thesis Space Domain Awareness Using Radio Telescope-Based Radar Systems ; Supervisor - Marco Martorella

10/08/2018 – 20/10/2020 Delhi, India

MASTER OF SCIENCE IN PHYSICS Department of Physics and Astrophysics, University of Delhi

- Studied core Master's-level subjects including Astronomy and Astrophysics, General Relativity, building a strong foundation in modern physics and space science
- Developed experimental, analytical, and scientific writing skills through laboratory work and technical report preparation

Website www.du.ac.in | **Final grade** 69.4 % | **Level in EQF** EQF level 7 |

Thesis Astrophotography of stars using a CCD camera in the observational astronomy lab

01/07/2015 – 01/07/2018 Delhi, India

BACHELOR OF SCIENCE (HONOURS) IN PHYSICS SGTB Khalsa College, University of Delhi

- Completed foundational coursework in Classical Mechanics, Mathematical Physics, Electromagnetic Theory, and Analog and Digital Electronics, establishing a strong base in theoretical and applied physics
- Developed analytical problem-solving skills and fundamental electronics knowledge through coursework and laboratory experiments

Field of study Physics | **Final grade** 7.392/10 | **Level in EQF** EQF level 6

WORK EXPERIENCE

PHD STUDENT – CNIT RASS LABORATORY – 01/11/2022 – Current – PISA, ITALY

- Conduct advanced research in satellite tracking, orbit determination, and radar-based sensing, with a focus on real-world space surveillance applications
- Develop and validate data fusion algorithms while contributing to academic publications and international collaborations

VISTING PHD STUDENT – CURTIN UNIVERSITY – 15/07/2025 – 14/10/2025 – PERTH, AUSTRALIA

• Developed and applied digital beamforming techniques for array-based sensing and surveillance applications
• Strengthened expertise in beamforming, passive radar architectures, and signal processing for Space Domain Awareness

RESEARCH FELLOW – INDIAN INSTITUTE OF REMOTE SENSING (IIRS), INDIAN SPACE RESEARCH ORGANISATION (ISRO) – 21/12/2021 – 31/10/2022 – DEHRADUN, INDIA

Website: https://www.iirs.gov.in/

• Worked on GNSS reflectometry for geophysical parameter retrieval
• Gained hands-on experience in spaceborne remote sensing and real satellite data analysis

PUBLICATIONS

2025
Role of Radio Telescopes in Space Debris Monitoring: Current Insights and Future Directions

Authors: Bhaskar Ahuja, Luca Gentile, Ajeet Kumar, Marco Martorella | **Journal Name**: Sensors | **Publisher**: MDPI

2025
Physics Aware Neural Networks for Satellite Tracking

Authors: Bhaskar Ahuja, Luca Gentile, Marco Martorella | **Journal Name**: Radio science Letters | **Volume, Issue and Pages**: Vol 7 | **Publisher**: URSI

2025
Entropy-Based Bayesian Model Averaging for Enhancing Classification Resilience Against Adversarial Perturbations Using SAR Images

Authors: Amir Hosein Oveis, Bhaskar Ahuja, Alessandro Cantelli-Forti, Marco Martorella | **Journal Name**: 2025 URSI Asia-Pacific Radio Science Meeting (AP-RASC), Sydney, Australia | **Publisher**: IEEE

2024
Advancements in Space Object State Estimation During Constant Thrust Maneuver Using Distributed Sensor Networks: A Radar Based Approach

Authors: Bhaskar Ahuja, Luca Gentile, Marco Martorella | **Journal Name**: 2024 International Radar Conference (RADAR) | **Publisher**: IEEE

2024
Improving Satellite Position and Velocity Estimation During Low-Thrust Maneuvers Using Multi-Bistatic Radar and the Unscented Kalman Filter

Authors: Bhaskar Ahuja, Luca Gentile, Marco Martorella | **Journal Name**: International Conference on Space Situational Awareness (ICSSA), Florida, USA

2024
A Novel Data Fusion Algorithm to Improve the Detection and Tracking of “Killer” Drones in Urban Environment

Authors: Walter Matta, Bhaskar Ahuja, Ajeet Kumar, Alessandro Cantelli-Forti | **Journal Name**: Proceeding of ICT Innovations 2024 International conference NATO Workshop

SKILLS

Digital skills

MATLAB | Python | Scientific Writing | Video Editing | Microsoft Office

Other skills

Team Management & Team Work | Scientific Communication | Working Under Pressure

Software Skills

Global Mission Analysis Tool (GMAT)

HONOURS AND AWARDS

09/05/2024
Best Paper Award at 4th IAA Conference on Space Situational Awareness (ICSSA) – International Academy of Astronautics (IAA)

Received the 3rd Best Student Paper Award for " *Improving Satellite Position and Velocity Estimation During Low-Thrust Maneuvers Using Multi-Bistatic Radar and the Unscented Kalman Filter* " presented during the 4th IAA ICSSA 2024 at Embry-Riddle Aeronautical University, USA.

18/03/2020
Graduate Aptitude Test in Engineering (GATE) -Physics – Indian Institutes of Technology Delhi

All India Rank 491

14/09/2019
National Eligibility Test (NET) - Physics – CSIR-UGC

All India Rank 12

10/05/2018
National Graduate Physics Examination (Top 1 %) – Indian Association of Physics Teacher (IAPT)

PROFESSIONAL ACTIVITIES

Academic Service

Reviewer — IEEE Transactions on Aerospace and Electronic Systems
Reviewer — IEEE Transactions on Radar Systems
Reviewer — European Radar Conference (EuRAD)
Reviewer — IEEE Radar Conference

LANGUAGE SKILLS

Mother tongue(s): **HINDI**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	B2	B2	B2	B2	B2
ITALIAN	A2	A2	A2	A2	A2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

HOBBIES AND INTERESTS

Mono-acting and theatrical performance

Fitness Training

CERTIFICATES

2024
Certification of participation in the conference on “Dynamics and Physics in the Solar System" at the Mathematics Department of the University of Pisa

2023
Certification of participation in the International Summer School on "Compressive sensing in Electromagnetism" organized by University of Trento, Italy

2023
Certification of participation in the 14th International Summer School on Radar/SAR organized by Fraunhofer-FHR, Germany

Autorizzo il trattamento dei miei dati personali presenti nel CV ai sensi dell’art. 13 d. lgs. 30 giugno 2003 n. 196 - “Codice in materia di protezione dei dati personali” e dell’art. 13 GDPR 679/16 - “Regolamento europeo sulla protezione dei dati personali”.

Pisa, Italy , 25/03/2026

BHASKAR AHUJA